

1. (Fall 2024, #8) Use results from the course to show that $f(x) = x^3 + 25x + 8$ has exactly one real root by showing the following steps. If a theorem is required, make sure to briefly, but explicitly, verify the hypotheses of the theorem.

(a) Show that $f(x) = x^3 + 25x + 8$ has at least one real root.

(b) Show that $f(x) = x^3 + 25x + 8$ has at most one real root.

2. (Spring 2024, #5) Consider the function over the interval $(0, 2)$ defined by:

$$f(x) = \frac{1}{x} - \frac{1}{x-2}$$

- (a) What are $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$?

- (b) Show that f is concave up over the interval $(0, 2)$.

- (c) Use an appropriate theorem to show that $f'(x) = 1$ has at least one solution over the interval $(0, 2)$.

- (d) Use part (b) to show that $f'(x) = 1$ has at most one solution in the interval $(0, 2)$.

3. (Spring 2023, #5) Consider the equation $e^x \cdot x^2 = 1$.

(a) Show that the equation has at least one solution on the interval $[0, \infty)$ and justify your work by citing the appropriate theorem(s) from this course.

(b) Show that the equation has at most one solution on the interval $[0, \infty)$ and justify your work by citing the appropriate theorem(s) from this course.

4. (Fall 2022, #4)

(a) Show that the equation $\cos(x) + 3x = 0$ has exactly one root.

(b) Find the absolute maximum and absolute minimum of $f(x) = \frac{x}{(x^2 - x + 1)}$ as x ranges over $[0, 3]$.

Hint: $f'(x) = \frac{(1-x)(1+x)}{(x^2 - x + 1)^2}$

5. (Fall 2021, #5) Find the exact number of solutions to the equation $4 - x = e^x$. Justify your conclusion by citing appropriate theorem(s) from this course.